Reflection and Traceability Report on Physiotherapy Companion

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1 Symbols, Abbreviations and Acronyms

symbol	description
FR	Functional Requirement
NFR	Non-Functional Requirement
HA	Hazard Analysis
MG	Module Guide
MIS	Module Interface Specification
SRS	Software Requirement Specification
TA	Teaching Assistant
VnV	Verification and Validation

[Reflection is an important component of getting the full benefits from a learning experience. Besides the intrinsic benefits of reflection, this document will be used to help the TAs grade how well your team responded to feedback. Therefore, traceability between Revision 0 and Revision 1 is and important part of the reflection exercise. In addition, several CEAB (Canadian Engineering Accreditation Board) Learning Outcomes (LOs) will be assessed based on your reflections. —TPLT]

2 Changes in Response to Feedback

2.1 SRS and Hazard Analysis

The hazard analysis document was improved through the peer review along with TA feedback in several areas for enhancement, including clarifying the roadmap with implementation timelines and rationales for safety requirements (#144), adding justification for future work prioritization (#150), and removing problematic assumptions about user familiarity with smart devices (#136). Inconsistencies between the SRS and Hazard Analysis were resolved (#129), while the system boundaries and components section was expanded with lower-level explanations to remove ambiguity (#135). The FMEA table was further decomposed to represent detailed failure modes rather than general effects (#142), and hazard categorization was clarified for technical hazards that could impact safety (#153). Additional improvements included adding rationales for each critical assumption (#136), addressing inconsistencies between assumptions and failure modes (#143), and noting where mitigation strategies would be expanded (#152). TA feedback further reinforced the need for roadmap rationales, which was addressed alongside similar peer feedback. One comment regarding bibliography completeness (#151) was determined to be invalid as the citation and reference were already properly included.

Source	Issue #	Issue Description	Changes Made	Commit
Peer Review	#151	Citation "IppolitoWallace1995" appears but bibliography incomplete	No change - comment invalid as bibliography and in-text citation were complete	None

Peer Review (Group 14)	#144	Unclear description of how/when safety requirements and hazard testing will be implemented	Added section in roadmap outlining rationale for accepting requirements within capstone timeline and rationales for excluded safety requirements	939b502
Peer Review (Group 14)	#150	Safety requirements prioritized as "first" vs "in the future" without clear justification or timeline	Added clear rationale for future work prioritization	470F490
Peer Review (Group 14)	#136	Missing rationale for each assumption; assumption about user familiarity with smart device should be avoided	Removed assumption about user familiarity; added rationales for each assumption	a103812
Peer Review (Group 14)	#143	Inconsistency between assumption about user familiarity and "confusing prompts" failure mode	Removed the initial assumption; will update language of "confusing prompts"	None (different commit)
Peer Review (Group 14)	#129	Inconsistencies between SRS and Hazard Analysis; hazards need prioritization by severity	Updated consistency between SRS and HA	Reference #410 PR

Peer Review (Group 14)	#135	System components need lower-level explanation with sample hazards; authentication module lacks detail	Added changes to HA system boundaries, splitting section to remove ambiguity	4bd898a
Peer Review (Group 14)	#142	FMEA rows should represent detailed failure modes rather than effects	Included further decomposition of table within FMEA	Different commit
Peer Review (Group 14)	#153	Technical hazards could also be safety hazards leading to unsafe exercise	Clarified hazard categorization	4381828
Peer Review (Group 14)	#152	Document lacks detailed mitigation strategies and implementation specifics	Mitigation included; will add section about effectiveness	None (different section)
TA	N/A	For Section 8 (Roadmap), including rationale would be helpful for which safety requirements to implement and why they are in that time frame	Updated the section with timelines and rationale for why each safety requirement was placed in each term	Similar to #144

Table 1: Feedback and changes for SRS and Hazard Analysis

2.2 Design and Design Documentation

2.3 VnV Plan and Report

3 Extras

The extras set out for the system consist of both a Design Thinking Document and a Usability Report. The Design Thinking Document aims to explain the process behind, and work done towards, the final design of the system. This document explains the overall flow of processing through the system and the interfacing elements, external libraries utilized, and more. The document flow is iterative, as is the design process, and aims to communicate the entire design process of the system. The Design Thinking Document can be found at docs/Extras/DesignThinking/DesignThinkingWorksheet.pdf.

The Usability Report aims to focus on evaluating the system from a user perspective. This report documents usability testing conducted with participants to assess ease of navigation, clarity of instructions, and overall user experience. It includes observations, user feedback, and analysis of interaction patterns to identify the key areas to focus on and address. The findings are used to ensure the system meets the non-functional requirements and to guide improvements to the interface for both patients and physiotherapists. The Usability Report can be found at docs/Extras/UsabilityTesting/UsabilityReport.pdf.

4 Design Iteration (LO11 (PrototypeIterate))

[Explain how you arrived at your final design and implementation. How did the design evolve from the first version to the final version? —TPLT]

[Don't just say what you changed, say why you changed it. The needs of the client should be part of the explanation. For example, if you made changes in response to usability testing, explain what the testing found and what changes it led to. —TPLT]

5 Design Decisions (LO12)

[Reflect and justify your design decisions. How did limitations, assumptions, and constraints influence your decisions? Discuss each of these separately. —TPLT]

6 Economic Considerations (LO23)

Yes, there is a clear market for our product. Existing physiotherapy companion apps such as Physiapp and Physiomobile offer progress tracking (checklist-based) and gamification (streaks), but they lack real-time exercise form feedback. Our app fills this gap by using OpenCV and MediaPipe pose landmarks to analyze patient movement during home exercises, giving immediate corrective feedback. Additionally, we provide value back to the physiotherapist by automatically sharing tracked metrics (e.g., reps, form quality) together with subjective patient data (pain levels, perceived exertion). This addresses the gap for patients and physios between appointments.

Marketing Approach

Marketing would target physiotherapy clinics and independent practitioners. Strategies include:

- Direct outreach to private clinics via email and informational webinars.
- Free trial periods for clinics to demonstrate improved patient adherence and outcomes.
- Testimonials and case studies from early-adopter clinics.

Cost to Produce Application

We estimate the cost to develop our application to a production-ready version (iOS/Android app, backend dashboard for physios, pose-analysis pipeline) at:

- Development (4 months, 2-3 engineers): $60,000 - $80,000.
- Exercise pose analysis expertise (consultation with physiotherapists to validate exercise models and feedback logic): $10,000 - $15,000.
- Cloud infrastructure (AWS/GCP for pose inference): $5,000 initial setup + $500/month.
- Legal/regulatory (privacy, medical device classification, liability for recording and providing feedback): $15,000 - $20,000
- Marketing launch: $5,000.

Total initial cost: approximately $95,000 - $125,000.

Pricing Model

We would not charge individual patients. Instead, we adopt a subscription model for physiotherapy clinics, tiered by the number of active patients per month. A "small private clinic" typically has 2-3 physiotherapists, each seeing 25-50 patients per week. Therefore, reasonable patient tiers are:

- **Tier 1 (up to 50 active patients / month):** $50 / month
- **Tier 2 (up to 150 active patients / month):** $100 / month
- **Tier 3 (unlimited patients):** $200 / month

Breakeven Analysis

Assume an average subscription of $100 / month per clinic (weighted toward Tier 2). Monthly recurring costs (cloud, support) are roughly $500 for every 100 clinics. Development cost is a one-time $110,000.

Monthly profit per clinic after variable costs: $100 - $5 = $95. Number of clinic-months needed to recover development cost:

$$\frac{\$110,000}{\$95 \text{ per clinic-month}} \approx 1,158 \text{ clinic-months.}$$

If we acquire 100 clinics in the first year, breakeven occurs after about 11.6 months. In terms of total clinics, we would need roughly 97 clinics subscribing for one full year to break even.

Potential Users (TAM / SOM)

- **Total Addressable Market (TAM):** All physiotherapists in Canada - approximately 22,000 registered physiotherapists (Canadian Institute for Health Information). Assuming an average of 5 physiotherapists per private clinic (typical for a medium-sized clinic), that gives roughly 4,400 clinics.
- **Serviceable Available Market (SAM):** Private clinics in Ontario, where our initial launch would focus. Ontario has roughly 6,000 physiotherapists, translating to about 1,200 private clinics.
- **Serviceable Obtainable Market (SOM):** In the first two years, with targeted marketing, we could realistically capture 1% of the Ontario market - about 12 clinics. That would generate monthly revenue of $1,200 (at $100 average). With successful promotion and marketing of the application, the application could grow to 5% (60 clinics) after three years, achieving monthly revenue of $6,000 and approaching a breakeven point.

7 Reflection on Project Management (LO24)

[This question focuses on processes and tools used for project management. —TPLT]

7.1 How Does Your Project Management Compare to Your Development Plan

[Did you follow your Development plan, with respect to the team meeting plan, team communication plan, team member roles and workflow plan. Did you use the technology you planned on using? —TPLT]

7.2 What Went Well?

In terms of what went well throughout our project, the team's overall organization and collaboration is one of the first few things that come to mind. We utilized GitHub Issues and version control to stay on track of all our deliverables, tasks, and progress. This ultimately allowed us to stay aligned and ensure everyone was held accountable for their assigned tasks. We also clearly divided up the respective responsibilities between the frontend and backend components to ensure everything remained as efficient as possible. Moreover, we successfully implemented real-time feedback, one of the core features of the app. We were able to successfully process frames continuously and return the relevant metrics (e.g., knee angle, rep count, ROM). This took numerous iterations to refine but we were so happy to see everything starting to finally come together.

7.3 What Went Wrong?

One of the main issues we faced was handling the accuracy of the pose detection and real-time motion analysis. Accurately detecting knee movement within a live environment was more difficult than expected, especially when dealing with inconsistent lighting, camera positioning, and partial visibility of the user. In addition to that, another issue we kept coming across was the accuracy of repetition counting. Any small variations in movement or added noise during the pose detection sometimes caused missed, delayed or incorrect rep counts. This required multiple iterations of threshold tuning and logic refinement, which took more time than initially planned. We also encountered numerous issues during the frame processing integration. Since the system relied on the backend for continuous frame processing, this sometimes led to performance issues such as latency or delayed feedback, which posed as a key limitation for us. Looking back, some of these technical challenges caused extreme delays and even though we continued to communicate as a team through these hard times, this did affect our pre-planned development schedule. However, regardless of these challenges, this whole experience including both the successes and challenges was all around a valuable learning experience for all of us, one that we will continue to hold very close to our hearts and are grateful for. Moreover, it genuinely helped us better understand the challenges of building real-time, user-based applications, which ultimately has better prepared us for our futures in our post-undergraduate careers.

7.4 What Would you Do Differently Next Time?

[What will you do differently for your next project? —TPLT]

8 Ethics and Equity (LO18)

[Describe how your team applied the principle of equity and universal design to ensure equitable treatment of all stakeholders. —TPLT]

9 Reflection on Capstone

[This question focuses on what you learned during the course of the capstone project. —TPLT]

9.1 Which Courses Were Relevant

Maham
SFWRENG 2AA4 - Software Design 1 introduced me to viewing software development through a more holistic lens. It raised my awareness of and emphasized many important areas such as what an MIS is, what software design principles are, and how to define software components mathematically.
SFWRENG 2DM3 and 2FA3 - Discrete Math 1 and 2 equipped me with the ability to translate natural language to mathematics. Given the MIS's extensive use of discrete math, proficiency in this area was imperative.
SFWRENG 3A04 - Large System Design introduced software architectural approaches to development. As part of a team, I had to develop a software and it was very similar to what was done in capstone, but at a significantly faster pace. This gave me valuable insight into what capstone entails.
SFWRENG 3RA3 - Requirements was essential to learning how to construct requirements for software that fulfills software quality benchmarks while also being tailored for the correct audience.
SFWRENG 3S03 - Software testing provided a concise introduction to software testing which was fundamental to deliver a functional product by the end of development.
SFWRENG 4HC3 - Human computer interfaces was a very important course, since it emphasized designing software with a human-centered approach. It taught us develpment priorities and the importance of always keeping the audience in mind.

Vaisnavi
Requirements Engineering (3RA3)

- Taught us how to engineer and refine system requirements.
- Helped significantly in defining the functional and non-functional requirements

Software Architecture (3A04)

- Taught us how to design larger systems as independent, modular components, which allowed for separation of concerns within our various modules

Software Testing (3S03)

- Taught us how to conduct both unit and integration testing.
- Helped us verify the reliability of key features like pose detection and repetition counting.

Software Development (2AA4 & 2C03)

- Taught us the fundamentals of software development and best practices for building scalable systems.

Human Computer Interfaces (4HC3)

- Taught and provided us with the correct tools to design our user interface, ensuring it was user-centered and intuitive.
- This course also provided us with the knowledge and tools to allow us to successfully conduct usability testing and refine our mobile application based on the user feedback.

9.2 Knowledge/Skills Outside of Courses

[What skills/knowledge did you need to acquire for your capstone project that was outside of the courses you took? —TPLT]